

《线性代数》 期中试题

姓名_____

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一、计算下列各题 (本题满分 70 分, 每小题 10 分)

1. 计算 $D_{n+1} = \begin{vmatrix} a_0 & a_1 & a_2 & \cdots & a_{n-1} & a_n \\ -x & x & 0 & \cdots & 0 & 0 \\ 0 & -x & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & 0 \\ 0 & 0 & 0 & \cdots & -x & x \end{vmatrix}$.

2. 求方程 $\begin{vmatrix} a_1 & a_2 & a_3 & \cdots & a_n \\ a_1 & a_1 + a_2 - x & a_3 & \cdots & a_n \\ a_1 & a_2 & a_2 + a_3 - x & \cdots & a_n \\ \vdots & \vdots & \vdots & & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_{n-1} + a_n - x \end{vmatrix} = 0$ 的根.

3. 设 $A = \begin{pmatrix} 5 & -1 & 3 \\ 0 & -2 & 1 \\ 0 & 7 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & -3 \\ 4 & -5 & 6 \\ -7 & 8 & 9 \end{pmatrix}$, $C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & -5 & 0 \end{pmatrix}$,
求 $(BC^T - I)^T(AB^{-1})^T + [(BA^{-1})^T]^{-1}$.

4. 设 $A = \begin{pmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & -6 & 0 & -7 & 2 \end{pmatrix}$, 求矩阵 A 的列向量组的一个极大无关组, 并用极大无关组
线性表示列向量组中其它向量.

5. 设 $A = \begin{pmatrix} 4 & 2 & 1 \\ 5 & 3 & 2 \\ 3 & 2 & 1 \end{pmatrix}$, A^* 为 A 的伴随矩阵, 求 $(A^*)^{-1}$.

6. 设 $\alpha_1, \alpha_2, \beta_1, \beta_2$ 都是 3 维列向量, 且 α_1, α_2 线性无关, β_1, β_2 线性无关. 证明存在非零向量 γ , 使得 γ 既可以由 α_1, α_2 线性表出, 也可以由 β_1, β_2 线性表出. 并在当 $\alpha_1 = (1, 0, 2)^T$, $\alpha_2 = (2, -1, 3)^T$, $\beta_1 = (-3, 2, -5)^T$, $\beta_2 = (0, 1, 1)^T$ 时, 求出所有的向量 γ .

7. 设有方程组 $\begin{cases} x_1 - 2x_2 + 3x_3 - x_4 + 2x_5 = 2, \\ 3x_1 - 5x_2 + 7x_3 + 2x_4 + 3x_5 = 5, \\ 2x_1 - 3x_2 + 4x_3 + 3x_4 + x_5 = a, \\ -x_1 + x_2 - x_3 - 4x_4 + x_5 = b, \end{cases}$ 试讨论 a, b 取何值时, 方程组有解, 并求解.

二、解答下列各题 (本题满分 30 分, 每小题 10 分)

8. 设方程组
$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0, \\ \dots\dots\dots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0, \end{cases}$$
 的解都是 $b_1x_1 + b_2x_2 + \cdots + b_nx_n = 0$ 的解, 证明: $\beta = (b_1, b_2, \dots, b_n)$ 是向量组 $\alpha_1 = (a_{11}, a_{12}, \dots, a_{1n}), \dots, \alpha_m = (a_{m1}, a_{m2}, \dots, a_{mn})$ 的线性组合.

9. 设向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关 ($s \geq 2$), 而

$$\beta_1 = \alpha_1 + k_1\alpha_s, \beta_2 = \alpha_2 + k_2\alpha_s, \dots, \beta_{s-1} = \alpha_{s-1} + k_{s-1}\alpha_s,$$

其中 k_i ($i = 1, 2, \dots, s-1$) 是任意常数, 证明: $\beta_1, \beta_2, \dots, \beta_{s-1}$ 线性无关.

10. 设方程组
$$\begin{cases} x_1 + a_1x_2 + a_1^2x_3 = a_1^3, \\ x_1 + a_2x_2 + a_2^2x_3 = a_2^3, \\ x_1 + a_3x_2 + a_3^2x_3 = a_3^3, \\ x_1 + a_4x_2 + a_4^2x_3 = a_4^3. \end{cases}$$

(1) 证明若 a_1, a_2, a_3, a_4 两两不等, 则此方程组无解.

(2) 设 $a_1 = a_3 = k, a_2 = a_4 = -k (k \neq 0)$, 且已知 β_1, β_2 是该方程组的两个解, 其中 $\beta_1 = (-1, 1, 1)^T, \beta_2 = (1, 1, -1)^T$. 试写出此方程组的通解.

《线性代数》参考答案

1.

$$D_{n+1} \xrightarrow{c_1+c_2+\dots+c_{n+1}} \begin{vmatrix} a_0+a_1+\dots+a_n & a_1 & a_2 & \dots & a_n \\ 0 & x & 0 & \dots & 0 \\ 0 & -x & x & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & x \end{vmatrix}$$

$$\xrightarrow{\text{展开} c_1} (a_0+a_1+\dots+a_n) \begin{vmatrix} x & 0 & 0 & \dots & 0 \\ -x & x & 0 & \dots & 0 \\ 0 & -x & x & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & x \end{vmatrix} = (a_0+a_1+\dots+a_n) x^n.$$

$$2. D_n \xrightarrow[i=2,3,\dots,n]{r_i-r_1} \begin{vmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ 0 & a_1-x & 0 & \dots & 0 \\ 0 & 0 & a_2-x & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & a_{n-1}-x \end{vmatrix} = a_1(a_1-x)(a_2-x)\dots(a_{n-1}-x).$$

原方程化为

$$a_1(a_1-x)(a_2-x)\dots(a_{n-1}-x) = 0.$$

若 $a_1 = 0$, 则一切 x 都为方程的解;

若 $a_1 \neq 0$, 则方程有 $(n-1)$ 个根:

$$x_1 = a_1, x_2 = a_2, \dots, x_{n-1} = a_{n-1}.$$

3. 先化简, 有

$$\begin{aligned} (BC^T - I)^T(AB^{-1})^T + [(BA^{-1})^T]^{-1} &= [AB^{-1}(BC^T - I)]^T + [(BA^{-1})^{-1}]^T \\ &= [AC^T - AB^{-1}]^T + [AB^{-1}]^T \\ &= CA^T. \end{aligned}$$

故

$$(BC^T - I)^T(AB^{-1})^T + [(BA^{-1})^T]^{-1} = CA^T = \begin{pmatrix} 10 & 0 & 0 \\ 5 & -2 & 5 \\ 10 & 10 & -35 \end{pmatrix}.$$

4. 记 $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5)$,

$$\text{对 } A \text{ 作初等行变换: } A \rightarrow \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

故 $\alpha_1, \alpha_2, \alpha_4$ 是列向量组的一个极大无关组, 且有 $\alpha_3 = 2\alpha_1 - \alpha_2 + 0\alpha_4$, $\alpha_5 = \alpha_1 + \alpha_2 + \alpha_4$.

5. 因 $(A^*)^{-1} = \frac{1}{|A|}A$, 又 $|A| = -1$, 故

$$(A^*)^{-1} = -A = \begin{pmatrix} -4 & -2 & -1 \\ -5 & -3 & -2 \\ -3 & -2 & -1 \end{pmatrix}$$

7. 对增广矩阵进行初等行变换:

$$\begin{pmatrix} 1 & -2 & 3 & -1 & 2 & | & 2 \\ 3 & -5 & 7 & 2 & 3 & | & 5 \\ 2 & -3 & 4 & 3 & 1 & | & a \\ -1 & 1 & -1 & -4 & 1 & | & b \end{pmatrix} \xrightarrow[r_3+r_1-r_2]{r_4-2r_1+r_2} \begin{pmatrix} 1 & -2 & 3 & -1 & 2 & | & 2 \\ 3 & -5 & 7 & 2 & 3 & | & 5 \\ 0 & 0 & 0 & 0 & 0 & | & a-3 \\ 0 & 0 & 0 & 0 & 0 & | & b+1 \end{pmatrix}$$

$$\xrightarrow{r_2-3r_1} \begin{pmatrix} 1 & -2 & 3 & -1 & 2 & | & 2 \\ 0 & 1 & -2 & 5 & -3 & | & -1 \\ 0 & 0 & 0 & 0 & 0 & | & a-3 \\ 0 & 0 & 0 & 0 & 0 & | & b+1 \end{pmatrix} \xrightarrow{r_1+2r_2} \begin{pmatrix} 1 & 0 & -1 & 9 & -4 & | & 0 \\ 0 & 1 & -2 & 5 & -3 & | & -1 \\ 0 & 0 & 0 & 0 & 0 & | & a-3 \\ 0 & 0 & 0 & 0 & 0 & | & b+1 \end{pmatrix},$$

故 $a = 3, b = -1$ 时, 方程组有解, 其通解为

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -9 \\ -5 \\ 0 \\ 1 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} 4 \\ 3 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad k_1, k_2, k_3 \in \mathbb{R}.$$

8. 令

$$A_{m \times n} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{nm} \end{pmatrix}, \quad B = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \\ b_1 & \cdots & b_n \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_2 \end{pmatrix},$$

由已知得到 $A\mathbf{x} = \mathbf{0}$ 与 $B\mathbf{x} = \mathbf{0}$ 同解, 则 $r(A) = r(B)$.

所以, B 中最后一行 $(b_1, b_2, \dots, b_n)$ 可由前 m 行线性表出.

9. 令

$$m_1\beta_1 + \cdots + m_{s-1}\beta_{s-1} = \mathbf{0},$$

即

$$m_1\alpha_1 + \cdots + m_{s-1}\alpha_{s-1} + (m_1k_1 + \cdots + m_{s-1}k_{s-1})\alpha_s = \mathbf{0}.$$

由 $\alpha_1, \dots, \alpha_s$ 线性无关, 知要使上式成立只能有

$$m_1 = \cdots = m_{s-1} = m_1k_1 + \cdots + m_{s-1}k_{s-1} = 0.$$

故 $\beta_1, \dots, \beta_{s-1}$ 线性无关.

另解. 由

$$(\alpha_1, \alpha_2, \dots, \alpha_s) \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & \vdots & \ddots & & \\ & 0 & 0 & \cdots & 1 \\ k_1 & k_2 & \cdots & k_{s-1} & 1 \end{pmatrix} = (\beta_1, \beta_2, \dots, \beta_{s-1}, \alpha_s),$$

注意到矩阵 $\begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & \vdots & \ddots & & \\ & 0 & 0 & \cdots & 1 \\ k_1 & k_2 & \cdots & k_{s-1} & 1 \end{pmatrix}$ 可逆, 故向量组 $\beta_1, \beta_2, \dots, \beta_{s-1}, \alpha_s$ 与向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$

等价, 从而向量组 $\beta_1, \beta_2, \dots, \beta_{s-1}, \alpha_s$ 线性无关, 得证其部分组 $\beta_1, \beta_2, \dots, \beta_{s-1}$ 也线性无关.

10. (1) 因 $i \neq j$ 时, $a_i \neq a_j$, $|\bar{A}|$ 为四阶范德蒙行列式, 故 $|\bar{A}| \neq 0$, 即秩 $(\bar{A}) = 4$, 而 A 为 4×3 矩阵, 秩 $(A) \leq 3$, 所以秩 $(A) \neq$ 秩 $(\bar{A})$, 题设方程组无解.

(2) 当 $a_1 = a_3 = k, a_2 = a_4 = -k (k \neq 0)$ 时, 得等价方程组

$$\begin{cases} x_1 + kx_2 + k^2x_3 = k^3, \\ x_1 - kx_2 + k^2x_3 = -k^3, \end{cases} \quad \text{即} \quad \begin{cases} x_1 + k^2x_3 = 0, \\ x_2 = k^2. \end{cases}$$

$$\bar{A} = [\bar{A} | b] = \left[\begin{array}{ccc|c} 1 & 0 & k^2 & 0 \\ 0 & 1 & 0 & k^2 \end{array} \right].$$

$r(A) = r(\bar{A}) = 2$. 基础解系只含 $n - r = 3 - 2 = 1$ 个解向量. 故

$$\alpha = \beta_1 - \beta_2 = [2, 0, -2]^T$$

为 $AX = 0$ 的解, 且为其基础解系, 故原方程组的通解为 $X = c\alpha + \beta_1$ 或 $X = c\alpha + \beta_2$ (c 为任意常数).